Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in>

Research Inquiry: Constrained RL for Lexicographic Arc Routing under Distribution Shifts

16 messages

Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Wed, Apr 1, 2026 at 12:35 AM
To: vaneet@purdue.edu

Dear Prof. Aggarwal,

I am Agriya, an undergraduate in Computer Science and Mathematics at Ashoka University, India. I am writing because your work on constrained and multi-objective reinforcement learning — particularly the primal-dual policy gradient methods for CMDPs and the multi-objective RL framework — connects to a structural challenge in my current project that I think you would find interesting.

I am developing ShARC, a framework applying risk-sensitive RL to the Hierarchical Capacitated Arc Routing Problem (HCARP). The problem involves a lexicographic makespan objective: minimize max(T1, T2, T3) across three priority classes of arcs, under simultaneous continuous distribution shifts on arc demands, traversal costs, and task availability. The lexicographic structure creates a multi-objective RL problem where the objectives are not simply weighted — they are ordered, with T1 dominating T2 and T2 dominating T3 in the optimization hierarchy. Your group's work on constrained MDPs with general policy parameterization (NeurIPS 2024) and on reinforcement learning for joint optimization of multiple rewards seems relevant to whether a principled policy gradient approach can handle this ordering without collapsing it to a weighted sum.

The distribution shift component adds a CVaR layer on top: the policy is trained to be robust in the worst-α fraction of shifted instances, which introduces a minimax structure over the lexicographic objective. I have not found work that addresses lexicographic objectives inside a CVaR minimax framework, and I suspect the interaction is non-trivial.

This is a faculty-supervised project at Ashoka targeting AAAI 2027. I have prior AAAI-affiliated publication experience (ICWSM). I would be very grateful for your thoughts on whether existing CMDP theory can accommodate the lexicographic ordering, or if this requires a new formulation. Happy to share a technical write-up.

Thank you for your time.

Best regards,
Agriya
B.Sc. CS & Mathematics, Ashoka University
linkedin
website

Vaneet Aggarwal <vaneet@purdue.edu> Wed, Apr 1, 2026 at 12:41 AM
To: Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in>

Our multi-objective uses a concave utility — and thus does not capture CVaR type metric.

Thus, I am not sure how you are planning to use this on lexicographic? Can you try giving a more detailed steps.

Best,
Vaneet

> On Mar 31, 2026, at 2:05 PM, Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> wrote:
>
> You don't often get email from agriya.yadav_ug2023@ashoka.edu.in. Learn why this is important
>
> ---- **External Email**: Use caution with attachments, links, or sharing data ----
>
> [Quoted text hidden]

Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Wed, Apr 1, 2026 at 12:52 AM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Dear Prof. Aggarwal,

Thank you for engaging with this — your question cuts to exactly the tension I have been working through, so let me be more precise.

You are right that CVaR is not a concave utility in the standard sense your framework assumes, and I overstated the directness of the connection in my initial email. Let me clarify what I am actually doing and where the genuine open question is.

The HCARP objective is a scalar: max(T1, T2, T3). Since this is a scalar episode return, CVaR_α[max(T1, T2, T3)] is well-defined as a standard tail risk measure over the distribution of that scalar induced by the shift distribution. So the CVaR layer is not in tension with the lexicographic structure per se — it operates on the output of the lexicographic aggregation, not inside it.

The place where your CMDP framework becomes relevant is in how I handle the lexicographic objective during training. Strict lexicographic minimization of (T1, T2, T3) is non-smooth and not amenable to direct policy gradient. The practical approximation I am considering is to relax it to a constrained MDP: minimize T1 subject to T2 ≤ β2 and T3 ≤ β3, where β2, β3 are budget parameters. This is a CMDP with two constraints, and your primal-dual policy gradient methods (particularly the zero-constraint-violation results) seem directly applicable to this relaxation.

The open question I was gesturing at — perhaps unclearly — is whether CVaR of the constrained objective can be optimized jointly with the constraint satisfaction guarantees. Concretely: in the CMDP relaxation, the primal-dual approach ensures the constraints T2 ≤ β2 and T3 ≤ β3 are satisfied in expectation. But under distribution shifts, some episodes will violate the constraints in the tail — and CVaR captures precisely that tail behavior. So the question is whether constraint satisfaction can be guaranteed not just in expectation but in CVaR, and whether the primal-dual structure survives that strengthening.

Is this a version of the problem your framework addresses, or does the CVaR constraint satisfaction requirement take it outside what is currently tractable?

Apologies for the imprecision in the initial email, and thank you again for pushing on this.

Best regards,
Agriya
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Vaneet Aggarwal <vaneet@purdue.edu> Wed, Apr 1, 2026 at 4:08 AM
To: Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in>

Sure — if we move to this type of constrained setup, where will the novelty be?
[Quoted text hidden]

Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Wed, Apr 1, 2026 at 10:11 AM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Dear Prof. Aggarwal,

The CMDP relaxation (minimize T1 subject to T2 ≤ β2, T3 ≤ β3) would be the optimization backbone, not the contribution. The novelty comes from three places that I believe fall outside existing work:
First, the theoretical object: your primal-dual framework guarantees constraint satisfaction in expectation across episodes. My setting asks for something strictly stronger — constraint satisfaction in CVaR, i.e., guaranteed in the worst-α fraction of episodes under distribution shift. To my knowledge, there are no results characterizing when primal-dual methods achieve CVaR-level constraint satisfaction rather than expectation-level. That gap is the theoretical contribution I am trying to make.

Second, the environment model: the simultaneous continuous shifts across arc demands, traversal costs, and task availability jointly produce a non-rectangular uncertainty set — the adversary cannot be decomposed into independent per-state perturbations. This is outside standard SA-rectangular and S-rectangular DR-RL theory, so even the

question of what guarantees are achievable here is open.
Third, the application: HCARP under simultaneous distribution shifts has not been studied with any RL method. The empirical contribution — demonstrating that a CVaR-constrained policy degrades more gracefully across shift severities than risk-neutral baselines — is independently novel regardless of the theoretical machinery used.

So to summarize: the CMDP relaxation gives us a tractable training algorithm. The novelty is in (a) the CVaR constraint satisfaction question, (b) the non-rectangular compound shift model, and (c) the routing domain. Does that framing make sense, or do you see the contribution differently?

Best regards,
Agriya
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Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Mon, Apr 6, 2026 at 9:43 AM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Hi Professor,

Just following up.
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Vaneet Aggarwal <vaneet@purdue.edu> Mon, Apr 6, 2026 at 7:42 PM
To: Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in>

Sure — seems good. Seems you already have the approach and problem, how can I help?

Vaneet
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Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Mon, Apr 6, 2026 at 8:09 PM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Dear Prof. Aggarwal,

Thank you — I appreciate you staying with this thread.

To give you something concrete: the specific place where I am stuck is the following. In the CMDP relaxation (minimize T1 subject to T2 ≤ β2, T3 ≤ β3), the primal-dual policy gradient approach guarantees constraint satisfaction in expectation across episodes. Under distribution shift, this expectation-level guarantee is over the training distribution — but at deployment, the shift distribution changes, and I cannot directly bound the probability that the constraints are violated in the tail.

The question I am trying to answer is: under what conditions on the uncertainty set (or on the policy class) does primal-dual training yield CVaR-level constraint satisfaction at deployment, rather than just expectation-level? I have not found a result that addresses this directly. The closest I can find is the safe RL literature on chance-constrained MDPs, but those results generally assume a fixed environment and do not account for distribution shift.

Specifically, I would be very grateful to know: (1) whether you know of any result in your group's work or adjacent to it that characterizes when primal-dual methods give tail-level rather than expectation-level guarantees, and (2) whether the non-rectangular compound shift structure I described earlier (jointly drifting demands, costs, and availability) makes this harder in a way that has a known characterization.

If it would help to have the formulation written up precisely, I am happy to share a draft of the problem formulation section — it is about 4 pages and has the MDP definition, shift model, and CVaR-REINFORCE estimator laid out formally.

Best regards,
Agriya
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Vaneet Aggarwal <vaneet@purdue.edu> Mon, Apr 6, 2026 at 11:26 PM
To: Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in>

Agreed — there are no tail level guarantees. CCV is measured in an average sense. We have something with peak constraints which may help: https://jmlr.org/papers/v24/21-0117.html

Best,
Vaneet
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Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Tue, Apr 7, 2026 at 5:15 PM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Dear Prof. Aggarwal,

Thank you — the peak constraints paper is very helpful, and I think it clarifies the theoretical gap more precisely than I had before.

As I understand it, the PCMDP framework (Bai, Aggarwal, Gattami 2023) gives the strongest possible guarantee: constraint satisfaction at every epoch with probability 1. Standard CMDPs give the weakest: satisfaction in expectation. What I need sits strictly between these — constraint satisfaction in CVaR, i.e., guaranteed in the worst-α fraction of episodes, under a shift distribution that changes between training and deployment.

The PCMDP paper does not account for distribution shift, and standard DR-RL results assume rectangular uncertainty sets. So the gap I am trying to work in is: can the peak constraint structure (or something weaker, like CVaR-level guarantees) be preserved when the environment shifts within a non-rectangular uncertainty set? I believe this is genuinely open, and the three-level hierarchy — expectation, CVaR, almost-sure — gives a clean framing for where the contribution sits.

I want to be direct: would you be willing to look at the problem formulation I have written up? It is about 4 pages long. If the gap I am describing is as clean as I think it is, I would be very grateful for your read on whether it is tractable enough to pursue as a theoretical result for AAAI 2027, and whether there is a proof strategy worth attempting.

Best regards,
Agriya
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Vaneet Aggarwal <vaneet@purdue.edu> Tue, Apr 7, 2026 at 7:19 PM
To: Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in>

Sure — happy to have a look.

Vaneet
[Quoted text hidden]

Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Tue, Apr 7, 2026 at 7:36 PM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Here it is, professor. I have also shared the literature review as a bonus if you want to go through it. Section 3 is the problem formulation which formally defines the base HCARP, sets up the non-rectangular compound distribution shift model across demand, cost, and availability , and casts the problem as a budget-augmented MDP for risk-sensitive RL.

Thank you for the continued engagement you have showcased through our conversation.
Thank you very much for your time and for agreeing to take a look.
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2 attachments

section3_sharc.pdf
238K

section6_sharc.pdf
163K

Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Mon, Apr 13, 2026 at 12:53 PM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Hi Professor,

Did you get a chance to go through the formulation perhaps?

Sincerely,
Agriya
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Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Fri, Apr 17, 2026 at 8:09 PM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Dear Prof. Aggarwal,

Following up on my email from 7th April with the problem formulation attached.
Happy to discuss at your convenience.

Sincerely,
Agriya
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Vaneet Aggarwal <vaneet@purdue.edu> Fri, Apr 17, 2026 at 8:32 PM
To: Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in>

This is a good problem. I have limited ideas to solve though.

Vaneet
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Agriya Yadav <agriya.yadav_ug2023@ashoka.edu.in> Fri, Apr 17, 2026 at 8:58 PM
To: Vaneet Aggarwal <vaneet@purdue.edu>

Alright professor, thank you for the continued engagement you have showcased nonetheless.

Sincerely,
Agriya
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